

Samshrita Ghimire

MTech Computer Science and Engineering **IIT Indore, India**

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Current Location: Indore, India

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Profile

Cybersecurity-focused graduate student with strong hands-on exposure to SOC operations, SIEM engineering, incident response, and vulnerability assessment, complemented by applied experience in machine learning and software fundamentals.

Education

M.Tech. in Computer Science and Engineering

2025 – Present

Indian Institute of Technology Indore, India

Research Focus: AI Security

CPI: **8.28 / 10.0**

Relevant Coursework: Machine Learning, Advanced Algorithms, Linear Algebra and Probability, Operating Systems, Compiler Design, Computer Architecture

Bachelor's in Computer Engineering

2019 – 2024

Tribhuvan University – Institute of Engineering, Nepal

Experience

Associate Cybersecurity Engineer

Oct 2024 – Jul 2025

Genese Solution, Lalitpur, Nepal

- Engineered and maintained **Wazuh SIEM**, including custom detection rules and alert tuning for SOC operations.
- Performed **incident triage and response** on real security events under supervised SOC workflows.
- Conducted **Vulnerability Assessment and Penetration Testing (VAPT)** for Class B and C banking systems.
- Executed **network-level analysis** using firewall logs and packet captures to support investigations.
- Carried out **dark web monitoring** using Cyfirma to identify potential credential and data leaks.
- Supported security compliance initiatives aligned with **ISO 27001** and Cyber Essentials frameworks.

Infrastructure Engineer Trainee

Sep 2024 – Oct 2024

FocusOne Payment Solutions (MOCO), Kathmandu, Nepal

- Assisted in deploying and securing servers, storage systems, and virtualized environments.
- Supported monitoring, performance optimization, and disaster recovery planning.

Cloud Security Engineering Apprentice

Dec 2023 – Aug 2024

Code Rush Pvt. Ltd., Lalitpur, Nepal

- Practiced Linux system hardening, secure user management, and shell scripting.
- Applied **OWASP Top 10** principles in web application security labs.
- Strengthened networking fundamentals for identifying and mitigating security risks.

Technical Skills

Cybersecurity & SOC Operations:

SIEM (Wazuh), Security Monitoring, Log Analysis, Incident Response, Detection Rule Engineering, Vulnerability Assessment, Penetration Testing (VAPT), OWASP Top 10, Threat Intelligence Basics, Dark Web Monitoring, Network Traffic Analysis (PCAP, Firewall Logs), ISO 27001 Fundamentals

Cloud & Systems Security:

AWS Fundamentals, IAM, Linux Security Hardening, Secure Server Configuration, Networking Fundamentals, Virtualization Basics

Machine Learning & Artificial Intelligence:

Supervised and Unsupervised Learning, CNN and LSTM Architectures, Computer Vision, Feature Engineering, Data Preprocessing, Model Training and Evaluation

Applied through academic coursework, projects, and hands-on training

Programming & Software Foundations:

Python, C, C++, SQL, Bash Scripting, Object-Oriented Programming, Data Structures and Algorithms (Coursework), Agile, Debugging, Git

Projects

Deepfake Video Detection System

Designed a CNN-LSTM based system to detect deepfake videos by analyzing temporal facial features, focusing on preprocessing, model evaluation, and robustness against manipulation-based attacks.

GitHub: github.com/samshrita/Deepfake-Video-Detection

Monument Recognition System

Developed a computer vision-based mobile application for real-time monument identification using object detection and classification techniques.

GitHub: github.com/samshrita/Monument-Recognition-System

Breast Cancer Detection System

Built a supervised machine learning model for early-stage breast cancer classification, emphasizing data preprocessing, feature selection, and performance evaluation.

GitHub: github.com/samshrita/Breast-Cancer-Detection

Certifications

- **CompTIA Security+** (Valid till 2027)
- **AWS Certified Cloud Practitioner** (Valid till 2027)
- **Cybersecurity, SOC, and Machine Learning Training** (Udemy)

Leadership & Activities

Board Member – Youth Echo Nepal

2024 – Present

- Led Internet safety and cybersecurity awareness programs across schools.
- Coordinated youth technology and entrepreneurship initiatives.

Secretary – Siru Youth Club

2023 – Present

- Organized community initiatives focused on education, environment, and youth well-being.

Awards & Scholarships

Homi Bhabha Scholarship (HJBSS)

2025 – Present

Awarded by the Ministry of Education, Government of India to pursue M.Tech. at IIT Indore.